

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) have demonstrated remarkable progress across a broad range of generative tasks, spanning image, video, and other high-dimensional synthesis settings. However, across these settings they still frequently produce hallucinated samples (*hallucinations*) that conflict with real-world knowledge, such as generating an implausible duplicate cup floating beside another cup. Despite how pervasively such failures arise, the absence of general and feasible methodologies for systematically quantifying hallucinations hinders progress on the problem as a whole and obscures broader pathways for designing next-generation generative models under factual constraints. In this work, we take a step toward closing this gap by studying a concrete and widely occurring form of hallucination, which we term **counting hallucination**: the generation of an incorrect number of instances or structured objects, such as a hand image with six fingers, despite such patterns being absent from the training data. As a general testbed for this phenomenon, we construct a dataset suite, **CountHalluSet**, equipped with well-defined counting criteria and spanning a spectrum of visual complexity through ToyShape, SimObject, and RealHand. Building on these datasets, we develop a standardized and broadly reusable evaluation protocol for quantifying counting hallucinations, and systematically examine how a wide array of sampling conditions in DPMs, including solver type, ODE solver order, sampling steps, and initial noise, shape counting hallucination levels. Furthermore, we analyze how these levels relate to commonly used evaluation metrics, such as the Fréchet Inception Distance (FID), revealing that widely adopted perceptual metrics fail to capture counting hallucinations consistently across settings.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, have achieved considerable success in generative modeling, producing high-quality samples across a wide and still-expanding range of modalities and tasks, including text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Despite these impressive advances, diffusion models across such domains still struggle to faithfully capture factual properties of the training data, and frequently generate outputs that violate consistency with real-world knowledge. A representative and recurring example is the generation of objects with an incorrect number of spatially intricate components, such as extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

A growing body of work has turned to identifying and analyzing common failure modes in diffusion models, particularly in the widely studied text-guided setting. For instance, Borji (2023) systematically catalogued several qualitative shortcomings shared across text-to-image models. Along similar lines, Liu et al. (2024) proposed an adversarial search method to uncover undesirable behaviors and failure cases in text-to-image generation, while Wu et al. (2023) employed an iterative approach to discover and mitigate spurious correlations.

More recently, *hallucination* has emerged as a distinct and increasingly prominent failure mode of diffusion models and has begun to draw broad attention across the research community (Aithal et al., 2024). Unlike the text-prompt-based failure modes discussed above, which can be assessed by measuring the alignment between input conditions and outputs (e.g., CLIP score (Hessel et al.,

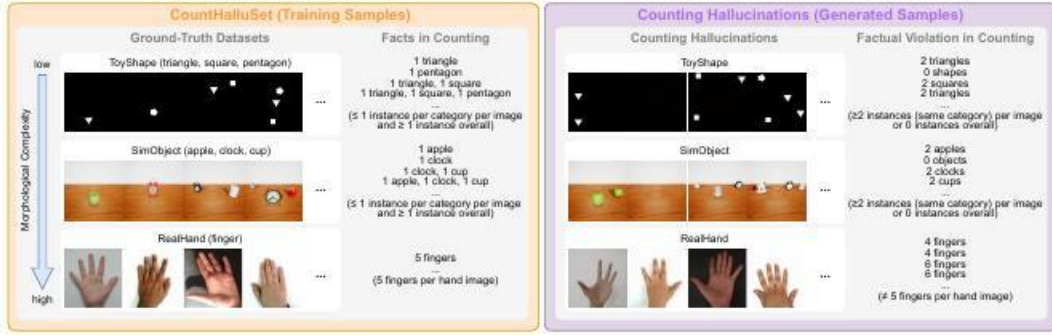


Figure 1: Illustrative comparison between the proposed **CountHalluSet** used for training (left) and the corresponding **counting hallucinations** generated by the model (right). Each dataset instantiates a different family of objects governed by its own counting criteria, together covering a range of settings. For example, in ToyShape and SimObject each image contains at most one instance per category and at least one instance overall, whereas in RealHand each image contains exactly five fingers. Counting hallucinations refer, in general, to generated images that violate a dataset’s counting rules, such as two apples in an image or an image containing no objects but a normal background, even though these patterns never appear in the SimObject dataset.

2021) and VQA score (Huang et al., 2025; Zarei et al., 2024)), or more conspicuous failures such as severe distortions, blur, and artifacts, hallucinated samples (*hallucinations*) tend to appear visually plausible and to arise largely independently of the conditioning signal. They manifest instead as subtle violations of factual consistency with respect to the training data, encompassing a range of phenomena such as incorrect object counts, geometries that look plausible yet are physically impossible, or other breaches of physical laws. Such fine-grained distinctions between valid samples and hallucinations pose a critical and general challenge for current AI systems, which frequently fail to detect them reliably. For example, even state-of-the-art large vision-language models like GPT-4o still struggle with seemingly basic tasks such as accurately counting the number of fingers in a clear hand image. For this reason, establishing a standardized and well-defined protocol for reliably quantifying hallucinated samples is essential: it lays a foundation for charting pathways toward next-generation generative models with strict factual consistency, a capability that is broadly critical for their role as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among the many types of hallucinated elements that arise in practice, such as counts, colors, spatial relationships, geometry, and physical realism, counting errors offer a particularly attractive entry point for systematic study, since they are at once easy to identify and straightforward to quantify, and thus serve as a clean proxy for the broader class of factual-consistency failures. This motivates us to begin by defining a concrete and general form of hallucination termed **counting hallucination**. *In this setting, the number of objects in the training data is explicitly predetermined. Consequently, a generated image is deemed hallucinated whenever its object counts deviate from these predefined values.* For example, an image depicting a hand with six fingers is classified as a counting hallucination, since such a pattern never occurs in the RealHand dataset, as shown in the last row of Fig. 1. To enable the study of this phenomenon across a range of conditions, we construct a dataset suite, CountHalluSet, comprising three datasets that together span a spectrum of morphological complexity for the countable objects: ToyShape (*triangle, square, pentagon*), SimObject (*mug, apple, clock*), and RealHand (*finger*). Figure 1 illustrates representative examples from these datasets alongside their corresponding counting hallucinations.

A critical step in our quantification protocol for counting hallucinations is the accurate identification and counting of objects. To this end, we employ counting models tailored to each dataset, an approach that generalizes naturally to any setting with well-defined counting criteria. Leveraging these datasets and counting models, we examine how a broad set of denoising conditions (e.g., sampling steps, initial noise, the order of ODE solvers, and solver type) within diffusion models affect counting hallucination rates. In addition, we conduct a correlation analysis between the counting hallucination rate and widely used perceptual metrics, including the Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021).

Our main findings and contributions can be summarized as follows:

- Numerical strategies that are commonly relied upon in diffusion models, such as increasing sampling steps from 25 to 100, often fail to mitigate counting hallucinations when generating human hands, even though the same techniques prove effective on simpler synthetic datasets, indicating that such heuristics do not transfer uniformly across generative settings.
- Across datasets and solver configurations, we consistently observe only a weak correlation between the counting hallucination rate and perceptual metrics, whereas a strong correlation emerges between the non-counting failure rate and the same metrics.
- Motivated by one of these findings, we propose a novel and broadly promising training paradigm, termed joint-diffusion models, which substantially reduces both counting-based hallucinations and non-counting failures on the RealHand dataset and points toward a general recipe for factual generation.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Inspired by the growing attention to hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), the study of hallucinations in diffusion models, including the unconditional setting, has attracted increasing interest. In particular, Aithal et al. (2024) proposed a qualitative hypothesis suggesting that hallucinated samples may arise from data interpolating between distinct modes. Despite this insight into potential causes, a systematic and generally applicable quantitative analysis of hallucinations in diffusion models remains lacking.

This motivates us to develop tools for the accurate identification and quantification of hallucinations in diffusion-generated outputs. Such an investigation can deepen our understanding of the undesirable behaviors these models exhibit, and clarify how the magnitude of denoising errors, such as variations in sampling steps or the order of ODE solvers, affects the severity of hallucinations across settings.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and the CLIP score (Hessel et al., 2021) are two widely used metrics for evaluating text-guided generated images. However, evaluation methods that rely on language-based models are inherently limited when repurposed to assess hallucinated images, and this limitation appears to be fundamental rather than incidental. One line of evidence is the statistical lower bound on the propensity of pretrained language models to hallucinate certain fact types, which caps the achievable detection accuracy of any purely language-based approach (Kalai & Vempala, 2024). A further indication is the persistent performance gap between human annotators and LLMs in classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023).

Beyond language-based measures, a family of quantitative evaluation metrics, such as FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021), is routinely employed across image generation tasks. These metrics build on feature embeddings extracted from a pretrained Inception-v3 model (Szegedy et al., 2016). Yet *whether such metrics can capture the severity of hallucinations in generated datasets* remains a broadly open question. In this work, we take a step toward answering it from the perspective of counting hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $x_0 \sim q_0(x_0)$ drawn from an arbitrary data distribution, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) introduce a *forward process* (a.k.a. the *diffusion process*) that gradually perturbs the data by adding Gaussian

noise over T discrete timesteps. At each timestep t , the marginal distribution of the noisy sample $\mathbf{x}_t$ conditioned on the original data $\mathbf{x}_0$ follows:

$$q_{0:t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_i\}_{i=1}^T$ is a pre-defined noise schedule dependent on t , which can take any of a variety of forms, with the linear (Ho et al., 2020) and cosine (Nichol & Dhariwal, 2021) schedules being common choices. The factor $\bar{\alpha}_t$ controls the signal-to-noise ratio at step t : $\bar{\alpha}_0 = 1$ (no noise) and $\bar{\alpha}_T \approx 0$ (nearly pure noise). An equivalent formulation of Eq. (1) is:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

As T becomes sufficiently large, the marginal distribution $q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ approaches a standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$, irrespective of the underlying data. In widely-used DPMs (Ho et al., 2020; Nichol & Dhariwal, 2021), T is typically set to 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (a.k.a. the *denoising process*) of DPMs aims to iteratively recover the observed data starting from the diffused data $\mathbf{x}_T$. Specifically, DPMs predict the noise $\boldsymbol{\epsilon}$ from the noisy data $\mathbf{x}_t$ using a neural network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$, known as the *noise prediction model*. The parameter θ is optimized by minimizing the following objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} [\pi(t) \|\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) - \boldsymbol{\epsilon}\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function.

Ancestral Sampling. Given noisy data $\mathbf{x}_T$, samples from the original distribution $q_0(\mathbf{x}_0)$ can be recovered by reversing the diffusion process over the same number of steps, a procedure commonly referred to as *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1 - \beta_t}} (\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the only unknown term, referred to as the *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function $-\sqrt{(1 - \bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ through the noise prediction model. Accordingly, an equivalent formulation of Eq. (3) is:

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1 - \beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \beta_t}} \frac{1}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. While the ancestral sampling in Eq. (4) contains a stochastic term (Kloeden et al., 1992), it admits an associated deterministic process that can be expressed as an ordinary differential equation (ODE), commonly known as the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau) \mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2} \beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2} \beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h(\mathbf{x}(\tau), \tau)$ and starts from an ideal diffused data $\mathbf{x}(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}(T))$. Set the discrete time steps as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, where $\tau_N = 0$. Applying the variation-of-constants formula for the exact solver over one step:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. Here $\Delta t > 0$ denotes the step size, which enables ODE-based solvers to accelerate sampling by taking larger steps than ancestral sampling, with the total number of sampling steps given by $T/\Delta t$. Since the linear part $\mathcal{G}(t_2, t_1)$ can be computed in closed form, the broad family of accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) concentrates its numerical effort, for instance through higher solver orders and more sampling steps, on accurately approximating the nonlinear integral term involving h . Further theoretical analysis can be found in Appendix A.1 through A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

In this section, we present a general methodology, together with the accompanying datasets, for quantifying counting hallucinations. We first give a dataset-agnostic definition of counting criteria and the associated counting hallucinations, and then introduce the datasets in the CountHalluSet suite as concrete instantiations of this framework. For each dataset, we specify its counting criteria and corresponding counting hallucinations, and outline the standardized protocol used for their quantification, a protocol designed to extend readily to other datasets with well-defined counts.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

In this section, we provide a general, dataset-agnostic definition of counting criteria within training datasets and characterize counting hallucinations as deviations from these predefined constraints. The formulation is deliberately abstract so that it applies to any dataset whose object counts can be specified.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ be an arbitrary training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the number of objects of category c in $\mathbf{x}$, denoted $N_c(\mathbf{x})$, is constrained to lie in a predefined set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), i.e., $N_c(\mathbf{x}) \in \mathcal{S}_c$. We further require that a valid reference sample contain at least one object, i.e., $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of categories under consideration.

Counting hallucinations as deviations from the counting criteria. Given any generated sample $\hat{\mathbf{x}}$, we say $\hat{\mathbf{x}}$ exhibits a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ if both of the following conditions hold:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ should exhibit sufficient visual quality that the background and any present objects are clearly discernible. This condition ensures that counting hallucinations can be cleanly separated from other prominent failure modes, such as severe degradations. We define a binary indicator $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, referred to as the *counting-ready indicator*, where 1 denotes that the image quality is sufficient for counting and 0 indicates otherwise. In practice this indicator can be realized in many ways, including human annotation, a classifier calibrated against human annotations, or other automated criteria, making the definition adaptable to a range of deployment settings.
2. **Violation of counting facts.** A violation of counting facts occurs if there exists a category c such that $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or if the image contains no objects (i.e., $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$), since the dataset requires at least one object per sample. In practice, we rely on a counting model to predict object counts in generated images as evidence for such violations, and any sufficiently accurate counter may serve this role.

A counting hallucination (CH) is identified if its quality is sufficient for counting ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and the predicted counts deviate from the counting criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluSet suite comprises three datasets that together cover a spectrum of settings for studying counting hallucinations, ranging from fully synthetic to real-world imagery: ToyShape, SimObject, and RealHand. Examples from each dataset and their corresponding counting hallucinations are shown in Fig. A1 and A2. Additional statistical summaries are provided in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. As the simplest instantiation in the suite, the ToyShape dataset comprises 30,000 images built from three geometric shapes: *triangle*, *square*, *pentagon*. All shapes are white, share an equal area of 120 pixels, and never overlap within an image. The counting criteria specify

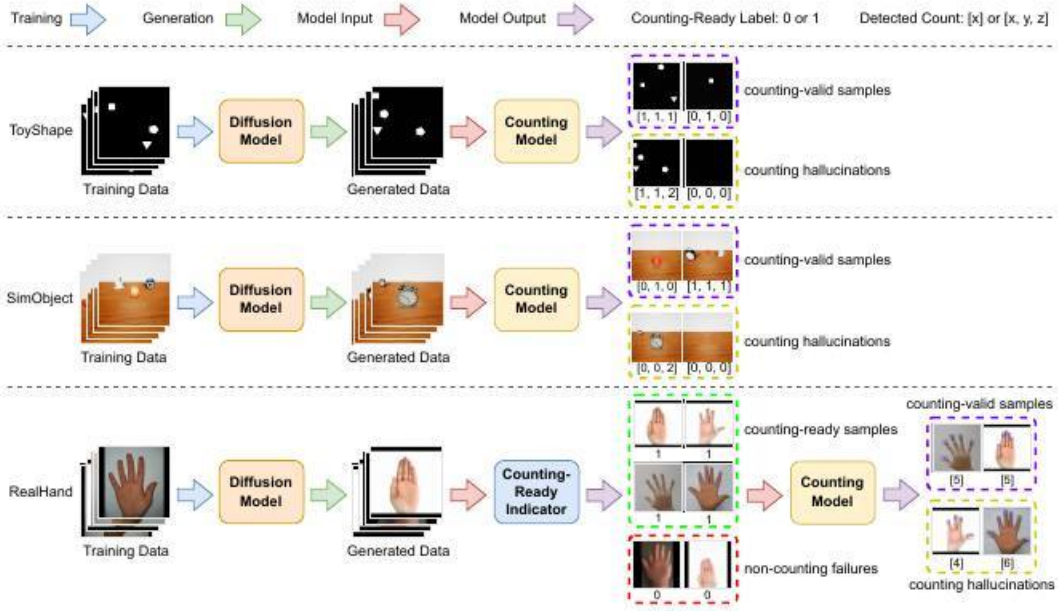


Figure 2: The general evaluation procedure used throughout the CountHalluSet suite. For the simpler ToyShape and SimObject settings, generated images are assessed for counting directly. For the more complex RealHand setting, a counting-ready indicator (a binary classifier aligned with human judgments) is additionally introduced to separate counting hallucinations from other non-counting failures, such as severely deformed fingers. Count labels correspond to ToyShape: *triangle*, *square*, *pentagon*; SimObject: *mug*, *apple*, *clock*; RealHand: *finger*.

that each image contains at most one instance of each shape category and at least one shape in total, as shown in the top-left of Fig. 1. Formally, we have $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$. For each class $c \in C$, we set $N_c(x) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(x) \geq 1$.

Counting hallucinations. Following the counting criteria defined above, any image generated by a model trained on ToyShape that contains two or more shapes of the same category is regarded as a counting hallucination, such as the image with two pentagons in the top-right region of Fig. 1. In addition, empty images (i.e., those showing only the black background without any shapes) are likewise classified as counting hallucinations. Further examples can be found in Appendix A.5.

Counting model used in ToyShape. We employ a counting model to identify counting errors within images, and forgo a counting-ready indicator here, since the dataset’s simplicity makes severely degraded images rare. Concretely, we fine-tune a ResNet-50 (He et al., 2016) on over 400k ToyShape images, with each category appearing 0–3 times per sample (i.e., $S_c = \{0, 1, 2, 3\}$), so as to cover a wide range of plausible scenarios spanning both counting-valid samples and counting hallucinations and thereby enable accurate quantification. The resulting model attains over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. Given a diffusion model trained on the ToyShape dataset, we generate a sample set matching the size of the training data (30,000 images). We then apply the counting model to predict the number of objects in each generated image. Finally, images that violate the counting criteria are flagged as counting hallucinations, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. Moving toward greater realism, the SimObject dataset contains 30,000 rendered images of everyday objects. It spans three object categories, *mug*, *apple*, and *clock*, each with 10 distinct intra-class variations. All objects are randomly placed on a wooden tabletop under fixed lighting, with little to no mutual occlusion, as shown in the middle of the left side of Fig. 1. The dataset is synthetically generated in Unreal Engine 5 (Epic Games, 2024), providing photorealistic rendering with faithful simulation of object geometry, material attributes, lighting, and reflections. The counting criteria mirror those of the ToyShape dataset (see Sec. 4.2.1). Formally, we have

$C = \{\text{mug}, \text{apple}, \text{clock}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Because the counting criteria closely parallel those of ToyShape, the definition of counting hallucinations, the construction of the counting model, and the quantification protocol all carry over directly, as shown in Fig. 2, illustrating how the same protocol applies unchanged across settings.

4.2.3 REALHAND

Dataset description. To probe counting hallucinations in challenging real-world scenarios, we construct the RealHand dataset, comprising 5,050 human hand images drawn from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². Specifically, we aggregate raw images from these diverse sources and discard those exhibiting blur or artifacts. Each image depicts a human hand with five fingers, and the collection spans both dorsal (back) and palmar (front) views under a variety of lighting conditions and backgrounds (e.g., white surfaces, blackboards, walls), as shown in the lower-left of Fig. 1. The counting criteria specify that each image contains exactly five fingers, i.e., $C = \{\text{finger}\}$ with $N_c(\mathbf{x}) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(\mathbf{x}) = 5$.

Counting hallucinations and non-counting failures. Generating realistic hand images is inherently susceptible to a broad range of unexpected failures beyond counting hallucinations (e.g., extra or missing fingers), including blurriness, severely deformed fingers, and various visual artifacts. To handle this richer failure landscape, we introduce a counting-ready indicator (i.e., a binary classifier) that identifies such cases. Examples of counting hallucinations and non-counting failure samples can be found in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To disentangle counting hallucinations from the failure modes noted above, we finetune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images to serve as a counting-ready indicator aligned with human judgments. For counting itself, we finetune a YOLO-12 (Tian et al., 2025) model on 2k fingertip-annotated generated images. Based on human evaluation of 500 randomly sampled images by three annotators, the indicator and counting model achieve over 96% and 99% accuracy, respectively. Examples of YOLO detection results can be found in Appendix A.6.

Quantification of counting hallucinations. Whereas ToyShape and SimObject permit direct evaluation of all generated images, owing to their simplicity and the rarity of severe failure cases (e.g., distorted or unrecognizable objects), the more demanding RealHand setting calls for an additional filtering step. Concretely, we first apply the counting-ready indicator to retain hand images of sufficient visual quality, and only then invoke the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. To span representative training regimes, we train a DDPM (Ho et al., 2020) from scratch for the ToyShape dataset and fine-tune latent diffusion models (LDMs) (Rombach et al., 2022), pretrained on the CelebA-HQ dataset (Karras et al., 2018), for the SimObject and RealHand datasets, thereby covering both training-from-scratch and pretrained-and-finetuned scenarios. The time step T is set to 1,000 with a standard linear noise schedule. Images are resized to 128×128 for DDPMs and 256×256 for LDMs. We optimize all models with Adam (Kingma, 2015) at a learning rate of 0.0001, running for 150k, 300k, and 80k steps on the ToyShape, SimObject, and RealHand datasets, respectively, with an effective batch size of 256.

Inference of Diffusion Models. For ancestral sampling (i.e., DDPM), the number of sampling steps is set to 1,000. For the ODE-based solvers, we cover both first- and second-order methods: DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)) as a first-order solver and DPM-Solver-2 (Lu et al., 2022) as a second-order solver. Each solver is evaluated under three widely-used denoising

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

Table 1: Counting hallucination rates (CHR; the number of counting hallucinations divided by the total number of generated samples), reported across a range of solver configurations and initial noise settings for all three datasets: ToyShape, SimObject, and RealHand. For RealHand, we additionally report the non-counting failure rate (NCFR), which captures failure cases identified by the counting-ready indicator. The total failure rate (TFR) is defined as the sum of CHR and NCFR. *Diffused* and *Normal* refer to ground-truth initial noise and standard Gaussian noise, respectively.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

step configurations, 25, 50, and 100 steps, so that our study spans the commonly encountered range of inference budgets. For the RealHand dataset, the counting model uses a YOLO detector with a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless noted otherwise, the number of generated samples matches the number of training samples in each evaluation, and all reported results are averaged over three random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports quantitative results on the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) across a broad set of denoising conditions, including solver types, ODE solver orders, and initial noise settings, evaluated on all three datasets: ToyShape, SimObject, and RealHand. Several general observations emerge.

(1) **Increasing sampling steps reduces counting hallucinations in synthetic datasets but exacerbates them in RealHand.** As shown in Table 1, under commonly used ODE solver settings (25, 50, and 100 steps), increasing the number of sampling steps generally lowers counting hallucination rates on ToyShape and SimObject, yet tends to raise it on RealHand, except at 100 steps with DPM-Solver-2. This contrasting behavior indicates that simpler synthetic distributions benefit from finer solver granularity, whereas more complex real-world distributions such as RealHand can behave in the opposite way, with additional steps overfitting local inconsistencies and thereby amplifying hallucinations, a divergence that cautions against assuming any single trend holds universally.

(2) **Employing higher-order ODE solvers effectively mitigates non-counting failures but exacerbates counting hallucinations in RealHand under the standard inference steps.** As shown in Table 1, comparing DPM-Solver-1 and DPM-Solver-2 on RealHand at standard sampling steps (e.g., 25 and 50 steps) reveals a consistent trade-off: the higher-order solver substantially reduces non-counting failures while simultaneously raising the counting hallucination rate. A similar pattern holds as the number of sampling steps grows for both DPM-Solver-1 and DPM-Solver-2. Taken together, these results suggest that allocating more computation (i.e., “better” sampling conditions) reliably improves overall perceptual quality (i.e., lower non-counting failure rates) but does not, in general, enforce the factual correctness of finger counts.

(3) **DDPM substantially outperforms ODE solvers in mitigating both counting hallucinations and non-counting failures.** Across all comparisons, ancestral sampling (i.e., DDPM) consistently attains the lowest CHR, NCFR, and TFR, indicating that DDPM provides a practical lower bound on the failure rates of generative models. More broadly, these results position ancestral sampling as an effective general strategy for minimizing hallucinations.

(4) **A better initial noise configuration can reduce both counting hallucinations and non-counting failures.** As shown in Table 1, using “diffused” noise (i.e., ground-truth initial noise) tends to yield lower CHR, NCFR, and TFR than “normal” noise (i.e., standard Gaussian noise). This effect likely arises because diffused noise better aligns with the initial noise distribution seen during training, suggesting more broadly that a more informed initialization can mitigate not only counting hallucinations but also other non-counting failures.

Table 2: Correlation of the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) with perceptual metrics, namely FID and MUSIQ, reported across datasets and solver families. Results for PTI-Hand are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the size of the RealHand dataset. FID for PTI-Hand is computed between RealHand images and the model-generated samples. Here, p denotes the p-value assessing the statistical significance of the correlation r or ρ . † indicates DDPM results integrated with those from various ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject		RealHand		PTI-Hand		RealHand		PTI-Hand		RealHand		PTI-Hand		RealHand		PTI-Hand		RealHand		PTI-Hand		RealHand	
	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p	r	p
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01	-0.98	<0.01	-	-	-	-
Pearson† (r)	0.49	0.14	-0.53	0.13	-	-	0.96	<0.01	-	-	0.46	0.20	-	-	-0.98	<0.01	-	-	-	-	-	-	-	-
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29	-	-	-	-	-	-
Spearman† (ρ)	0.75	0.01	-0.51	0.15	-	-	0.95	<0.01	-	-	0.40	0.28	-	-	-0.91	<0.01	-	-	-	-	-	-	-	-
Distance Correlation (r_d)	0.89	-	0.97	-	0.47	-	0.99	-	0.91	-	0.93	-	0.42	-	0.97	-	0.99	-	-	-	-	-	-	-
Distance Correlation† (r_d)	0.82	-	0.74	-	-	-	0.95	-	-	-	0.71	-	-	-	0.97	-	-	-	-	-	-	-	-	-

(5) Greater morphological complexity in countable objects increases the likelihood of counting hallucinations. Comparing CHR across ToyShape, SimObject, and RealHand reveals a clear trend: diffusion models exhibit higher CHR as the morphological complexity of the target objects grows. This points to a general difficulty, in that the fine-grained details and complex spatial relationships intrinsic to morphologically complex objects make it harder for diffusion models to preserve accurate object counts, leading to more frequent additions or omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Since counting hallucination is a form of factualness hallucination, it is natural to ask a broadly relevant question: *Does a better perceptual metric, such as a lower FID, necessarily imply fewer counting hallucinations by reflecting a smaller distribution gap between generated and training data?* Our answer is **not necessarily**.

To probe this relationship, we conduct a correlation analysis between CHR and perceptual metrics, and likewise between NCFR and the same metrics. We consider two widely used and complementary perceptual measures: a distribution-based metric, FID (Heusel et al., 2017), and an instance-based metric, MUSIQ (Ke et al., 2021). For correlation estimation, we employ three complementary statistics that together cover a range of dependence structures: the Pearson correlation (Lee Rodgers & Nicewander, 1988) (linear dependence), the Spearman rank correlation (Spearman, 1904) (monotonic dependence), and the Distance Correlation (Székely et al., 2007) (general nonlinear dependence). The following evidence supports our conclusion.

✚ **The correlations between CHR and perceptual metrics are dataset-dependent and solver-dependent rather than intrinsic.** As shown in the first and third rows of Table 2, the Pearson correlations between CHR and FID vary substantially from one dataset to another (positive for SimObject, negative for RealHand, and positive again for PTI-Hand), while the corresponding Spearman correlations shift from positive to negative to negative. The distance correlations likewise decrease from 0.82 (SimObject) to 0.74 (RealHand) to 0.47 (PTI-Hand), indicating that increasingly complex generative content progressively weakens the association between CHR and FID. Integrating DDPM results with those from ODE solvers attenuates these correlations even further, reinforcing that the relationship is not intrinsic but contingent on the setting.

✚ **The correlations between NCFR and perceptual metrics are consistently strong and positive.** In sharp contrast to the weak and setting-dependent correlations between CHR and perceptual metrics, the correlations between NCFR and perceptual metrics are strong and robust across all settings. For instance, every correlation coefficient exceeds 0.88 for NCFR vs. FID, with very small p-values indicating high statistical significance.

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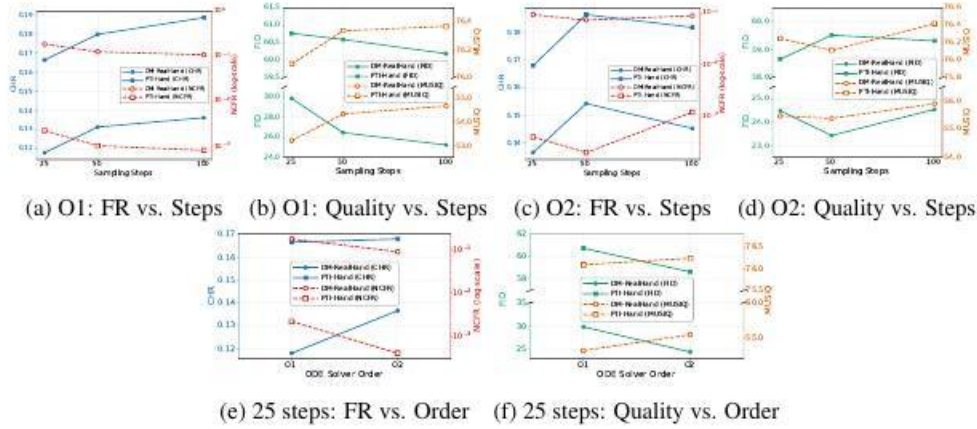


Figure 3: Comparison of trend consistency between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium), illustrating how the observed behavior generalizes across model scales. O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). FR metrics include CHR and NCNR; quality metrics include FID and MUSIQ. Across variations in sampling steps and solver order, both models exhibit highly consistent trends.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

To test how far our findings extend beyond the controlled setting, we evaluate counting hallucinations in a large-scale pretrained text-to-image model. Specifically, we assess Stable-Diffusion-3.5-Medium (Esser et al., 2024) using the very same counting-ready indicator and counting model introduced in Sec. 4.2.3, without task-specific modification. Additional details are provided in Appendix A.8. As shown in Figure 3, the manner in which solver type and sampling steps influence counting hallucinations in this large pretrained model aligns closely with the trends revealed by our controlled RealHand experiments. This consistency reinforces the validity of the findings in Sec. 5.2 and, more importantly, underscores the broad applicability of our evaluation protocol. The correlation results in Table 2 further support this conclusion: the same weak correlations between CHR and perceptual metrics, and strong correlations between NCNR and perceptual metrics, reappear in this large-scale regime. Together, these results demonstrate the robustness and generality of our findings and show that the proposed evaluation protocol transfers reliably from controlled settings to large-scale pretrained text-to-image models.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The lower counting hallucination rates observed for simpler countable objects (see Observation 5 in Sec. 5.2) suggest a general principle: diffusion models capture counting-based factuality more reliably when the underlying visual structure has low morphological complexity (e.g., masked objects). This raises an important and broadly applicable question: if a diffusion model were to denoise general visual representations jointly with explicit, simple structural constraints that are easier to model, such as object masks that encode basic topology and counts, within a shared latent space, could such a joint learning paradigm mitigate counting hallucinations? We term models that jointly learn both general visual representations and structural constraints during denoising **joint-diffusion models** (JDM), and view the idea as a general recipe instantiated here on hands.

Concretely, we perform a channel-wise concatenation of the latent representation of the hand mask with that of the corresponding RGB image, as shown in Fig. 4. We then finetune the UNet pretrained on CelebA-HQ (Karras et al., 2018) (following the RealHand configuration described in Sec. 5.1) on RealHand images together with their associated masks, which we extract using SAM-2 (Ravi et al., 2024). To accommodate both modalities, we expand the input and output channels of the denoising UNet to match the combined latent dimensionality of the image and mask latents (i.e., $3 + 3 = 6$ channels).

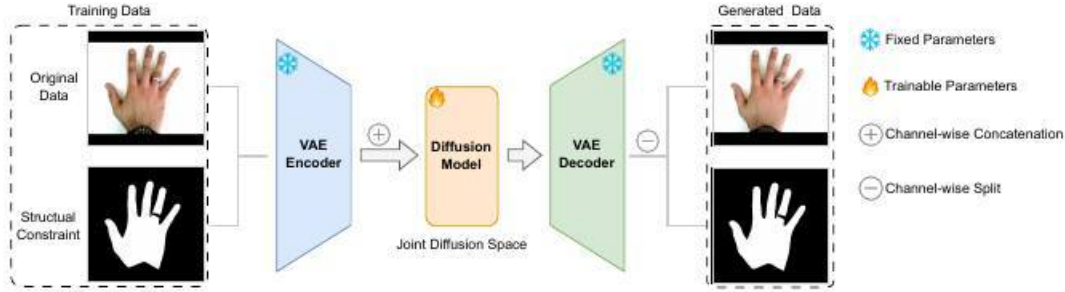


Figure 4: Diagram of the proposed joint-diffusion model (JDM), a general training paradigm illustrated here on hands. The model performs channel-wise concatenation of the latent representations of hand RGB images and their corresponding segmentation masks. This joint learning paradigm encourages anatomically plausible hand poses and promotes correct finger topology during generation, resulting in reduced CHR and NCFR.

Table 3: Comparison of failure rates between LDM and our proposed JDM on the RealHand dataset across a range of ODE solvers and sampling steps. Lower values indicate better performance. For JDM, although the model jointly generates hand RGB images and segmentation masks, FID is computed solely on the RGB outputs to ensure comparability with the baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fr�chet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

As shown in Table 3, JDM consistently attains lower failure rates than the vanilla LDM, on both CHR and NCFR, across a diverse set of sampling conditions, highlighting its effectiveness at faithfully generating structurally intricate objects that are prone to counting ambiguities. JDM does, however, yield higher FID. This degradation likely stems from the increased denoising burden of jointly modeling two modalities under a small-scale training regime, which forces the model to trade off perceptual quality for improved structural realism. Encouragingly, we also observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning settings, hinting that the paradigm scales beyond the controlled regime studied here. Further details, results, and discussion can be found in Appendix A.9.

6 CONCLUSION

In this work, we take a step toward systematically quantifying hallucinations in diffusion models broadly and offer new insights into improving factual generation across this field. Centering on counting hallucinations as a general and tractable case, we introduce **CountHalluSet** together with a practical, reproducible, and reusable methodology for measuring such hallucinations. Through comprehensive and controlled quantification experiments, we show that numerical strategies commonly relied upon do not, in general, mitigate counting hallucinations when generating human hands. Moreover, our correlation analyses demonstrate that widely used perceptual metrics (e.g., FID and MUSIQ) fail to consistently reflect the severity of counting hallucinations across settings. Experiments on large-scale pretrained text-to-image models further corroborate both the generality and the validity of our findings, and we expect the framework and observations to extend to a wider range of generative models and factual-consistency failures.

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A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the true reverse process $p_\theta(\mathbf{x}_{T:0})$, their KL divergence is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

For both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$, we can decompose them as follows, leveraging the Markov property:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

By applying the chain rule to decompose the KL divergence, we have:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

Then, according to the Data Processing Inequality (Cover & Thomas, 2012), marginalizing (or discarding some random variables) will not increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error can be obtained by combining Eq. (A.4) and Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality ($=$) holds strictly only in the ideal case where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$, meaning that the diffusion model recovers the underlying data distribution $q(\mathbf{x}_0)$ exactly. This upper bound on the overall KL divergence ($\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0))$) between true and model-generated data coincides, when the data entropy $H(\mathbf{x}_0)$ is excluded, with the variational bound derived in DDPM (Ho et al., 2020), which suggests in general that optimizing DDPM’s variational loss directly tightens this explicit upper bound on the discrepancy between true and model-generated data.

Diffused-prior gap. The first source of error arises from the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ and the prior distribution used for sampling $p_\theta(\mathbf{x}_T)$. This mismatch is a general consequence of the number of diffusion steps T being finite in practice. The discrepancy can be quantified formally using any suitable distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence. Any other appropriate metric, such as FID or the Wasserstein distance, could equally be employed to evaluate this distributional discrepancy. Present from the outset, the diffused-prior gap induces a deviation that persists throughout the denoising process and ultimately manifests as a sustained output error.

Noise model prediction error. The second source of error stems from the noise prediction model. Its estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ , as guaranteed in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ denote this prediction error. Based on Eq. (5), any discrepancy in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\alpha_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Since each reverse step builds on the mean from the previous iteration, this bias accumulates throughout the denoising trajectory and ultimately contributes to the overall prediction error.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Based on Eq. (7), we take a one-step numerical solver as a representative example, the analysis extending naturally to broader solver families. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and starts from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Similarly, let $\hat{\mathbf{x}}_k$ denote the trajectory computed by a numerical ODE solver at discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ be the estimated model function. For a given numerical trajectory from $\hat{\mathbf{x}}_0$ to $\hat{\mathbf{x}}_N$, we seek to characterize its total accumulated trajectory error. Let $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ be the trajectory error at step k , and consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Using the variation-of-constants formula for the exact solution over one step (from τ_k down to τ_{k+1}):

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. Note that the integration limits are backward in time.

Numerical path evolution. A one-step numerical solver approximates the solution. This numerical step deviates from the exact evolution governed by $\hat{h}$ starting from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ be the exact solution at τ_{k+1} of the ODE using $\hat{h}$, starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

So, we have:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtract the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12):

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. We can unroll this recurrence relation from $k = 0$ to $N - 1$. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Applying the recurrence

repeatedly:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial error term captures the deterministic contribution to the final path error of a given initial error instance, whose statistical origin lies in the distributional Diffused-Prior Gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As shown in Eq. (2), the forward process in DDPM (Ho et al., 2020) can be formulated as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

And the reverse process is defined as follows:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to that in Eq. (A.16), while its deterministic reverse process (with $\eta = 0$) is defined as follows:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share an identical forward (diffusion) process, DDIM does not merely strip the stochastic term from DDPM’s reverse (denoising) process. The distinction arises because the two methods adopt different coefficient formulations for the predicted noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from the three datasets used throughout this study. As reported in Table A1, each dataset maintains a balanced distribution across its respective categories. Table A2 further shows that the distribution of images with different object counts is comparable, indicating the absence of object-count bias across all datasets. The datasets appear in the same order in both tables, and the category order in Table A1 is defined as follows:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: Number of objects per category across each dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Number of images by object count(s) across each dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We provide representative examples of counting hallucinations spanning all three datasets, as shown in Fig. A2. For the ToyShape and SimObject datasets, counting hallucinations either contain more than two instances of at least one category or contain no instances at all while still showing a normal background, as shown in Fig. A2(a) and Fig. A2(b). In the RealHand dataset, counting-hallucinated samples are characterized by incorrect finger counts (e.g., 4 or 6 fingers), as shown in Fig. A2(c). Notably, the last two images of the first row in Fig. A2(c) appear at first glance to display five fingers; however, one finger is partially incomplete and visibly artifacted, effectively leaving four valid fingers. These samples are therefore classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative examples of YOLO predictions on the RealHand dataset. The YOLO detector reliably identifies fingertips across a wide variety of hand poses and orientations, spanning both dorsal and palmar views. It may nonetheless fail on anatomically implausible configurations, such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). Such counting-valid samples are still detected as five-finger hands and thus remain indistinguishable from anatomically valid ones in terms of finger counts.

Figure A3(b) shows several counting-hallucinated samples for which YOLO predicts incorrect fingertip counts (e.g., 3, 4, 6, or 7 fingertips). Figure A3(c) further illustrates several rare edge cases (<1%) in which the detector fails to identify fingertips with unclear boundaries, cracked fingers, or floating artifacts. Such rare cases can be partially mitigated by tuning YOLO’s inference hyperparameters (e.g., IoU and confidence thresholds), though doing so may degrade detection performance on the majority of counting-valid samples. Because a consistent detection setup is applied uniformly across all evaluations on the RealHand dataset, the presence of these rare cases does not affect the validity of our overall conclusions, even though fully resolving them remains a challenging and low-reward endeavor.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows representative examples of the three classes of generated hand images: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, the RealHand evaluation procedure first applies the counting-ready indicator to separate counting-ready samples (i.e., counting-valid and counting-hallucinated) from images unsuitable for counting. The latter typically exhibit severe visual artifacts, such as distorted, cracked, partially visible, or floating

fingers. Counting-ready samples are then further classified as counting-valid or counting-hallucinated according to whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we adopt 10 different hand-generation prompts to prompt Stable-Diffusion-3.5-Medium, chosen to cover a range of viewpoints and phrasings. The prompts are listed below.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

The definition of hallucinations assumes that the model generates patterns never observed during training. This assumption, however, is difficult to verify for large-scale pretrained diffusion models, whose full training distribution is inaccessible. To keep the quantification of counting hallucinations feasible in such general cases, we adopt a conservative criterion: we treat any generated hand image containing six or more fingers as a counting-hallucinated sample. This criterion is well justified, since while real images may show fewer than five visible fingers due to occlusion, articulation, or viewpoint, it is highly unlikely for real data to contain hands with six or more fingers. Finger counts ≥ 6 therefore provide a reliable and empirically verifiable signal of counting hallucination that transfers to settings where the training distribution is unknown. Generated examples can be found in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

In the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, as opposed to 80k steps for the vanilla LDM. This larger training budget is required because extending the latent space from three to six channels in JDM introduces a substantial mismatch with the pretrained latent distribution of LDM. Consequently, additional optimization is needed to close this distributional gap and allow JDM’s modeling capacity to fully materialize.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

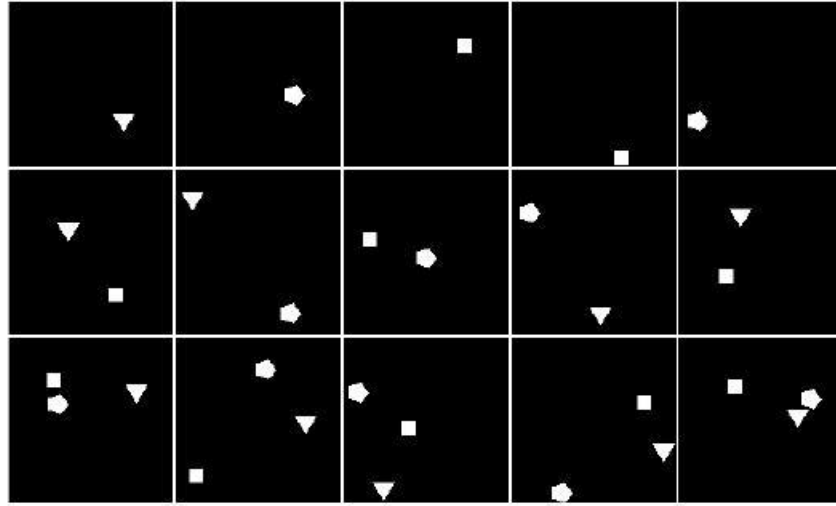
Importantly, JDM is best understood as a **novel training paradigm** for improving factual generation in diffusion models generally, rather than as a superior inference strategy. On the one hand, JDM extends only the input and output channels of the UNet while leaving all intermediate layer sizes unchanged, so it **introduces a negligible inference-time overhead compared to vanilla LDM**. On the other hand, augmenting a pretrained text-to-image model with additional mask channels **disrupts the manifold geometry of the pretrained latent space and undermines the original denoising distribution**, which makes directly finetuning such a model nontrivial.

Table A3: Comparison of failure rates and image quality between finetuning SD-1.5 and finetuning SD-1.5 with the joint-diffusion training paradigm on RealHand, illustrating how the paradigm behaves when applied to a large-scale pretrained model. The first row (baseline) reports results from finetuning Stable-Diffusion-1.5 on RealHand data. The second row reports results obtained by freezing all intermediate layers of the baseline model and finetuning only the input and output layers with extended channel dimensions. Lower values indicate better performance.

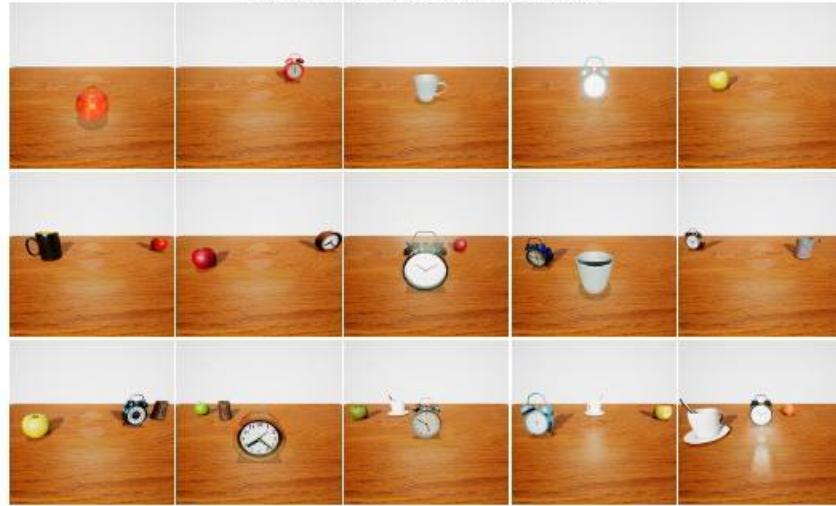
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

Despite this challenge, we still observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning conditions. In practice, we find that jointly aligning JDM’s extended latent space with the highly rigid latent geometry of large pretrained models is difficult given the limited scale of RealHand. To strike a practical balance, we adopt a two-stage finetuning strategy: we first finetune a Stable-Diffusion-1.5 model on RealHand, then freeze all intermediate UNet layers and finetune only the input and output layers with extended channels. As shown in Table A3, even under this highly constrained setup, the joint-diffusion paradigm consistently yields lower counting hallucination rates at 25 and 50 sampling steps across both solvers.

Overall, the evidence presented here indicates that JDM is a promising and broadly applicable direction for enforcing factual consistency in image generation. Finally, we believe that **exploring its behavior and performance under true large-scale training remains an important and compelling avenue for future research.**



(a) Examples in the ToyShape dataset

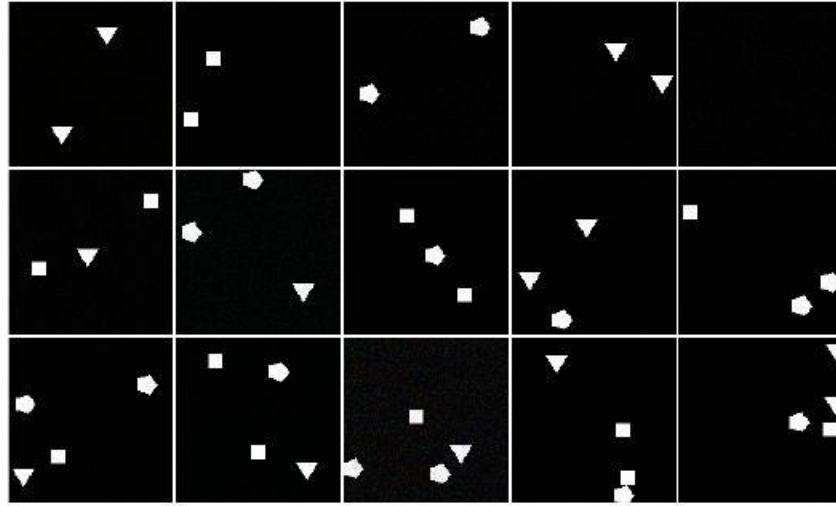


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

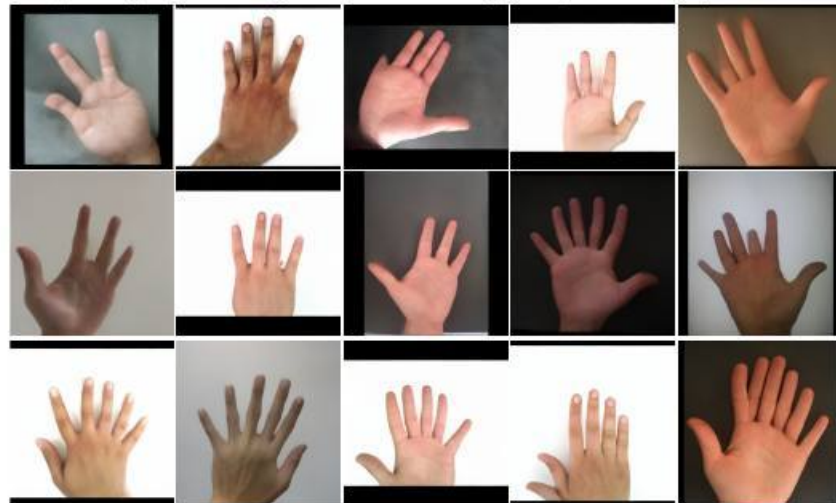
Figure A1: Representative examples spanning the datasets of the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

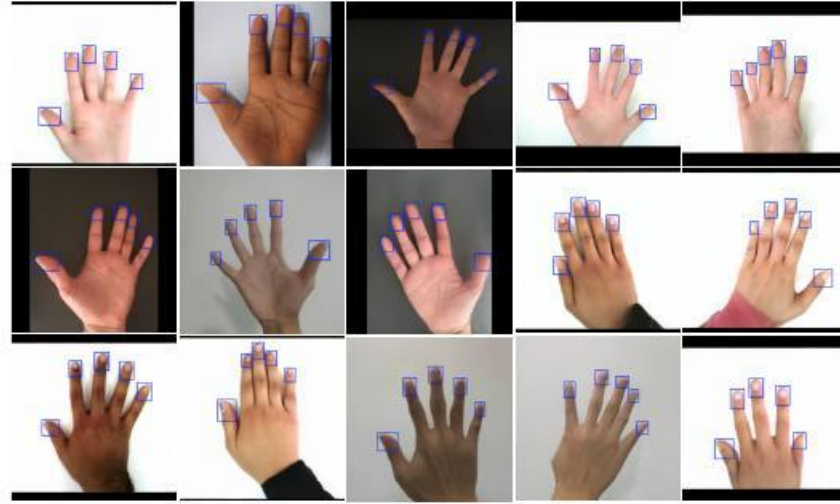


(b) Examples of counting hallucinations corresponding to the SimObject dataset

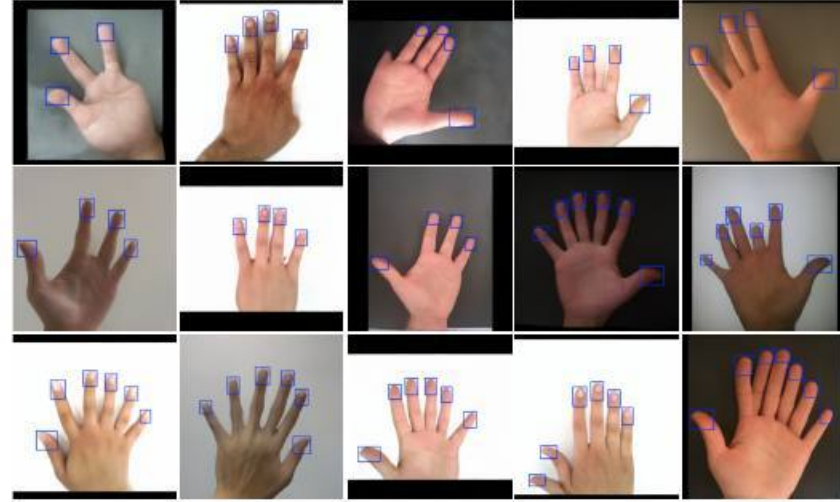


(c) Examples of counting hallucinations corresponding to the RealHand dataset

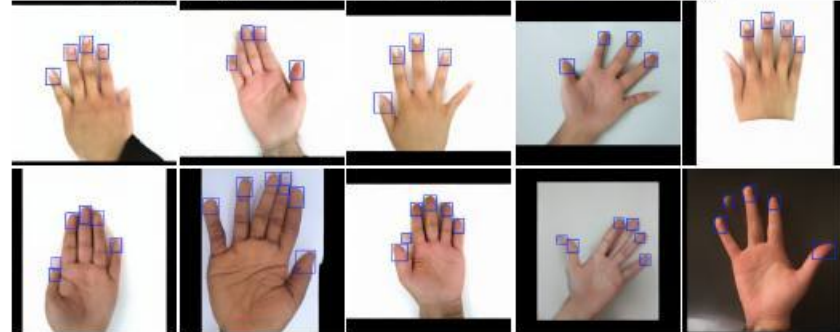
Figure A2: Examples of counting hallucinations spanning the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers



(b) Counting-hallucinated YOLO predictions with not five fingers



(c) Rare edge cases where YOLO predictions fail

Figure A3: Examples of YOLO predictions across generated RealHand images. (a) Counting-valid samples with five fingertip bounding boxes correctly predicted by YOLO (including cases with two thumbs). (b) Counting-hallucinated samples with incorrect fingertip count predictions (e.g., 3, 4, 6, or 7 detected fingertips). (c) Several rare edge cases that the counting-ready indicator fails to filter out and YOLO struggles to handle, such as blurred, incomplete, or anatomically disconnected fingers.

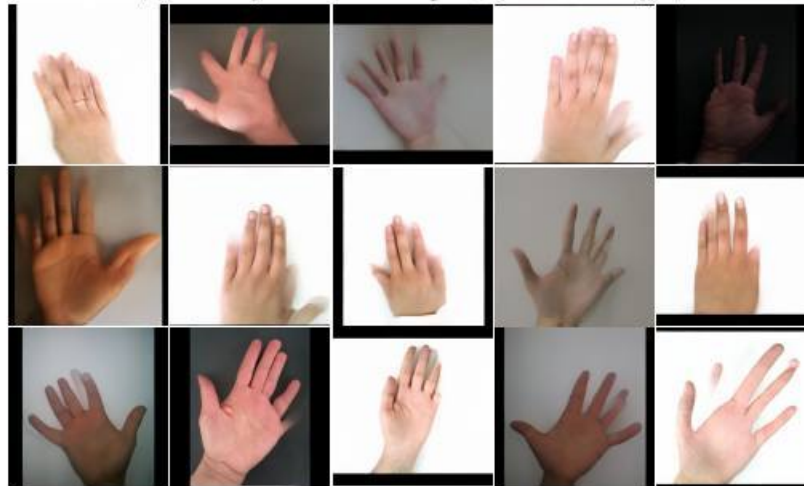
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(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: Examples of the three classes of generated hand images. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: Examples of the three classes of generated hand images produced by pretrained Stable-Diffusion-3.5-Medium (PTI-Hand), demonstrating that the same taxonomy carries over to a large-scale pretrained model. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with more than five valid fingers (e.g., 6 or 7). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.